

UNIVERSITY PARTNER



Human Computer Interaction
(5CS020)

Usability Report

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1. Introduction

The ease of utilizing and using certain things can be defined as usability. It is the metric to measure the easiness of utilization of any product. Usability is generally associated with fewer errors and stress with effective and efficient ways to interact with products such as they can be used to achieve a certain task. It is measured by success rate, time, error rate, and satisfaction (Coursera, 2024).

Usability in computer science evaluates the quality of the interface of a computer, specifically how easy the interfaces are. Usability in mobile applications emerges as efficiency, satisfaction, effectiveness, learnability, memorability, cognitive load, simplicity, and ease of use (Pawee, 2020).

Usability is crucial for:

- The user can control the application and know where they are on the application.
- Helping users such that it reduces short-term memory load and provides information from the interface
- The system that matches the mental model that the user has of the real world and should talk with user's language.
- The interface can communicate clearly and efficiently as much as possible (Stanford University, 2024).

2. Literature Review

Usability of mobile applications: literature review and rationale for a new usability model: The journal represents models of usability by Nielsen and from ISO (International Organization for Standardization). The usability factors are efficiency, learnability, memorability, satisfaction, and errors including utility. Likewise, ISO elaborates on usability as efficiency, satisfaction, emphasizes the user, the context of use, goal, and effectiveness. The journal introduces a new model for usability, unlike Nielsen and ISO as they were built majorly for traditional desktop applications causing issues of connectivity, small screen size, and limited processing power. Thus, using the PACMAD method can bridge the gap of cognitive overload in the usability of mobile devices due to

its deployment in multitasking and size constraints. So, usability cognitive load and mobile-related factors of mobile applications need to be considered in usability making it comprehensive (Harrison, et al., 2013).

Usability Dimensions for Mobile Applications-A Review: A set of usability dimensions that are specifically made for mobile application context due to the lack of comprehensive usability guidelines for mobile applications can be analyzed and made through twenty-five dimensions which consider user, environment, technology, and activity/task. It is important to consider the usability of mobile applications as the importance of mobile devices in various domains like business and logistics is increasing. As mobile users are to exceed 3.3 billion connections usability considering the diverse user demographics is necessary. As mobile phones have been used for various domains for logistics, disaster management, tourism, work related tasks considering it is vital for the mobile applications to not only be functional, technically nevertheless it is also required to be user-friendly thus, the usability as much important. The efficiency in several environments of mobile applications should be product capacity to be learned, understood and operated easily (Baharuddin, et al., 2013).

The products of mobile applications include: Netflix and LinkedIn

3. Analysis

3.1 Shneiderman's "8 Golden Rules of Interface Design":

3.1.1 Clarity

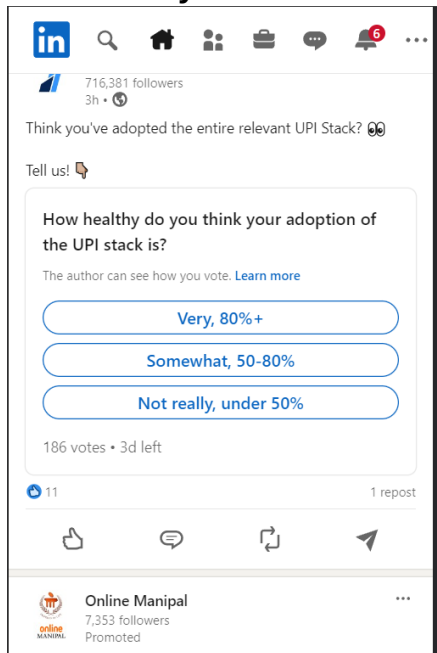
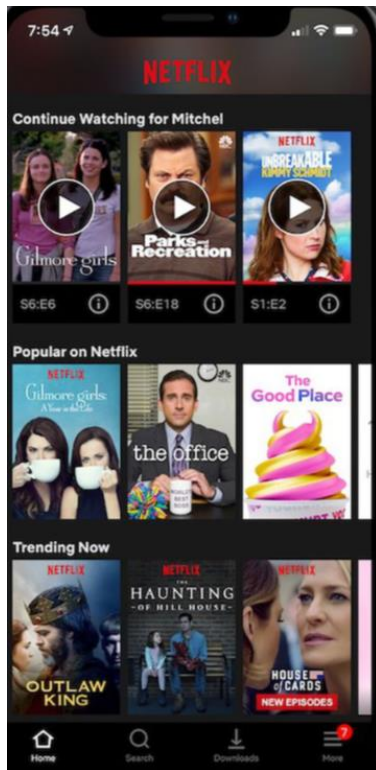


Figure 1Clarity in LinkedIn

As we can see, the LinkedIn app has a straightforward and a very clear layout. User's news feed is shown very concisely for users to see the updates. Navigation tabs are also clearly labeled.



*Figure 2*Clarity in Netflix

As we can see, Netflix app also has clarity as its top design presenting users with “Trending Now” and “Top Picks” where user can easily search for TV shows and movies.

3.1.2 Consistency:



Figure 3 Consistency in LinkedIn

As we can see, LinkedIn has consistency in design throughout the app. The icons, buttons and menus are consistent in appearance for smooth navigation.

The icons and overall design is very consistent as seen in the picture.

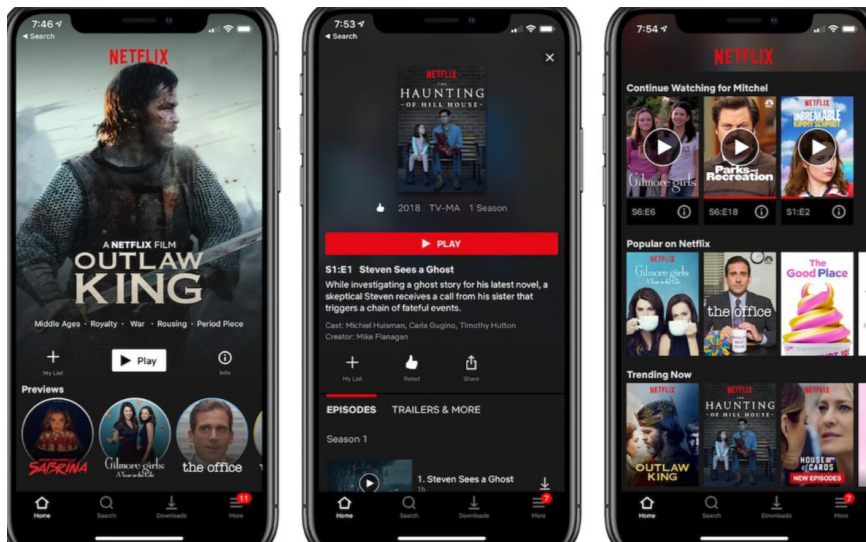


Figure 4 Consistency in Netflix

As we can see, Netflix also has consistency regardless of its viewing list, browsing contents.

Consistency can be seen compared to each and every pages.

3.1.3 Simplicity:



Figure 5 Simplicity in LinkedIn

As we can see, LinkedIn keeps the features simple and plain by emphasizing on the essentials which minimizes the clutter. Navigation too is straightforward which reduces confusion.

A simple design in LinkedIn:

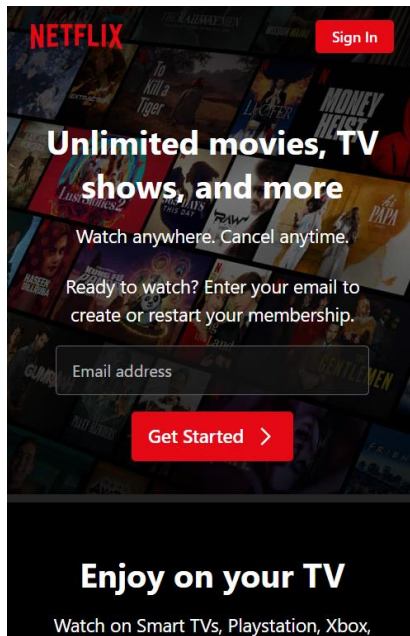


Figure 6 Simplicity in Netflix

As we can see, Netflix also adheres to this principle which in spite having a lot of movies and a lot of function the design is very clean and easy through which users can browse the app easily.

3.1.4 Feedback:

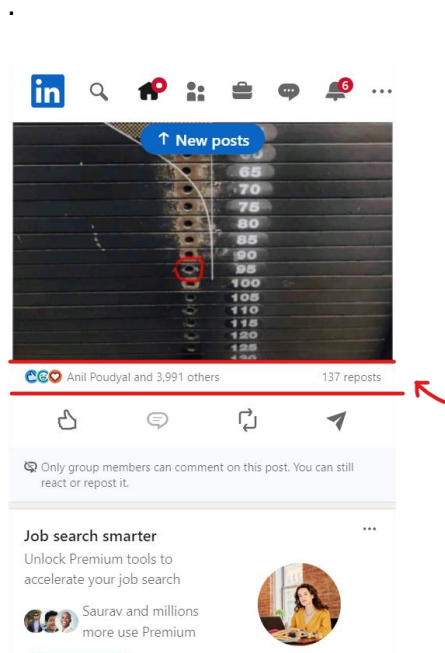


Figure 7 Feedback in LinkedIn

As we can see, LinkedIn displays the likes, comments also share counts on the posts as the user interacts and the user also receive notifications for messages and other activities which are relevant

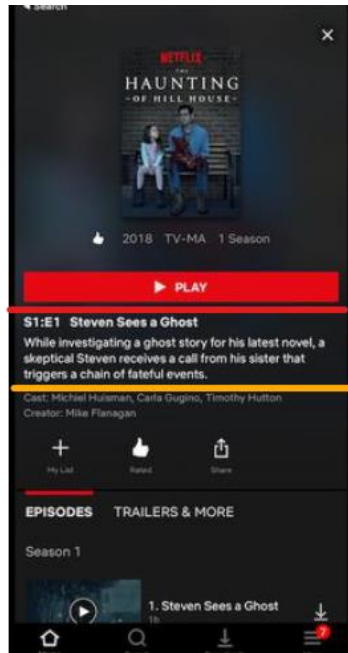


Figure 8 Feedback in Netflix

As we can see, Netflix offers the feedback to the user through ratings on a show, their reviews and a synopsis of every shows. They also receive feedback through playback controls and seek options.

3.1.5 Flexibility:

In iPhone 12 Pro:

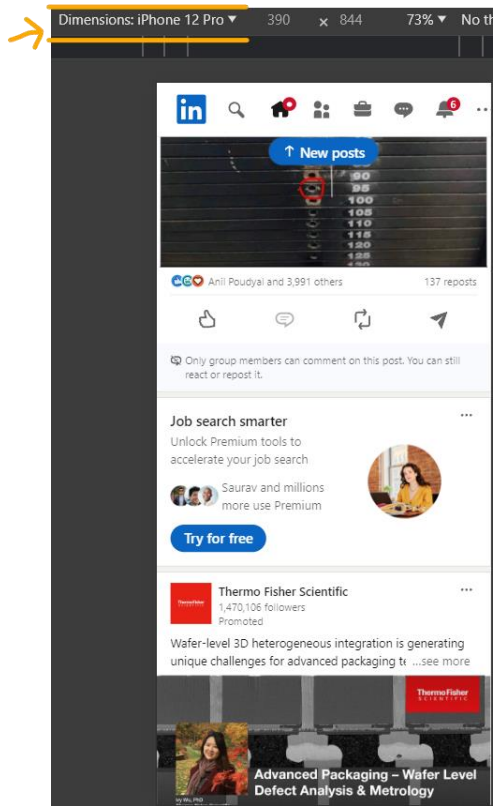


Figure 9 Flexibility in LinkedIn

In Samsung Galaxy S20 Ultra:

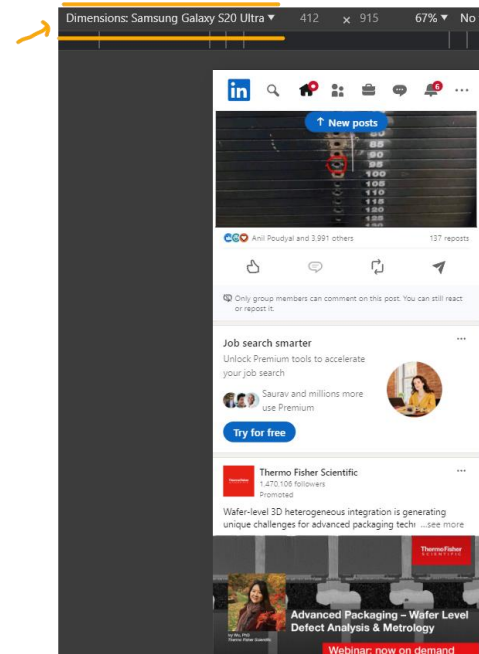
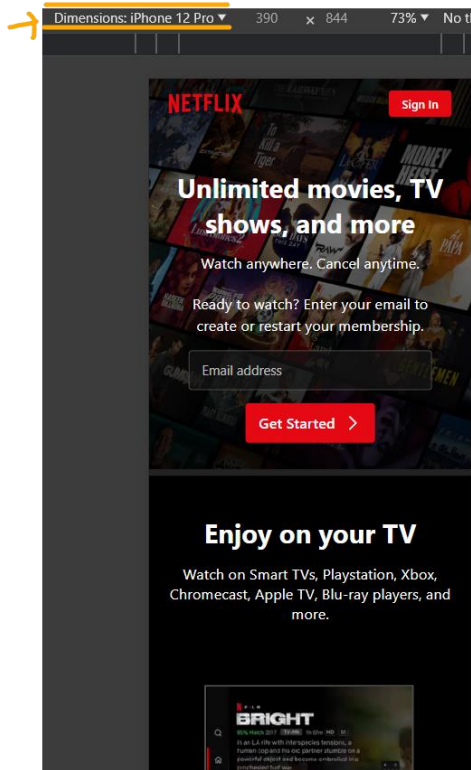


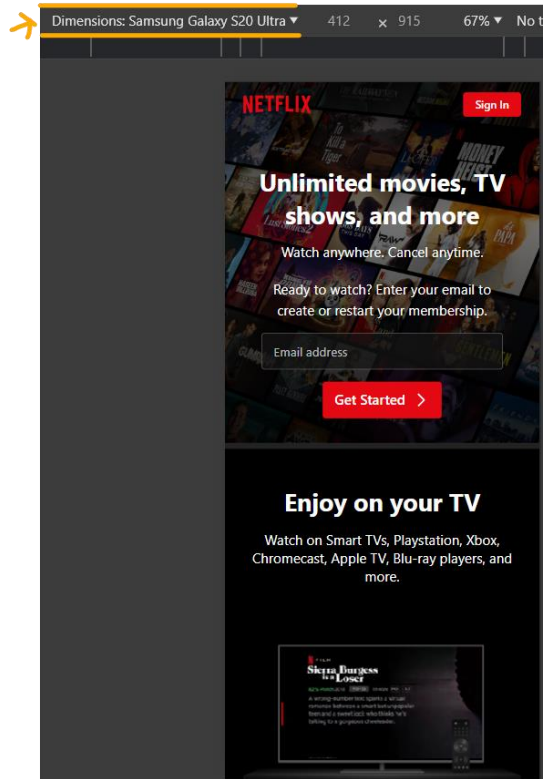
Figure 10 Flexibility in Netflix

As we can see, LinkedIn app adapts to vivid screen sizes and orientation for consistent user experience for many mobile devices.

In iPhone 12 Pro:



In Samsung Galaxy S20 Ultra:



As we can see, Netflix app supports for different screen for different mobile devices for flexibility for optimal user experience.

3.1.6 Visual Hierarchy:

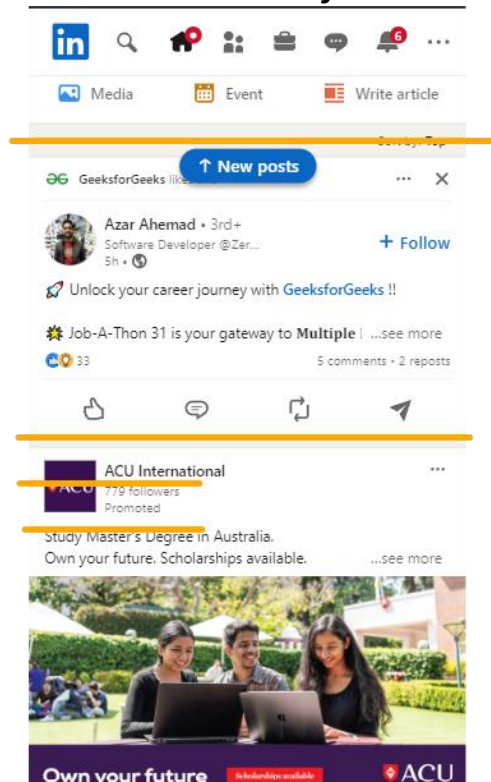


Figure 11 Visual Hierarchy in LinkedIn

As we can see, in LinkedIn contents are prioritized on the home feed whereby important updates are displayed prominently and other contents of suggested posts are presented later. First posts liked from connection is shown and then the promoted content is shown.

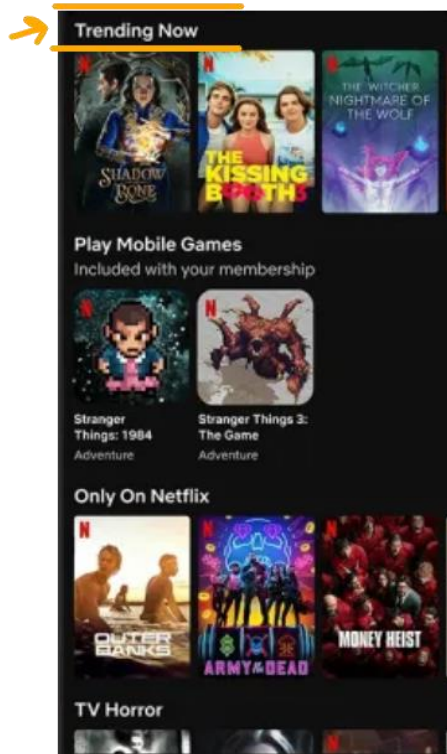


Figure 12 Visual Hierarchy in Netflix

As we can see, Netflix deploys visual hierarchy whereby they feature recognizable titles and high quality images to grab user's attention. Here they have shown the trending things which definitely would attract user's attractions.

3.1.7 Accessibility:

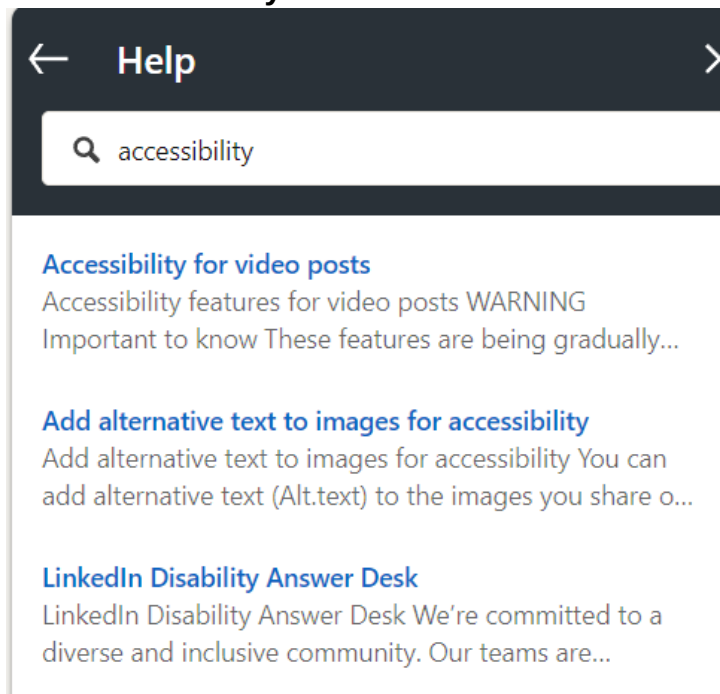


Figure 13 Accessibility in LinkedIn

As we can see LinkedIn provides with great accessibility features to add alternative text to images, accessibility in video post, answer desk, adjust text size and contrasts.

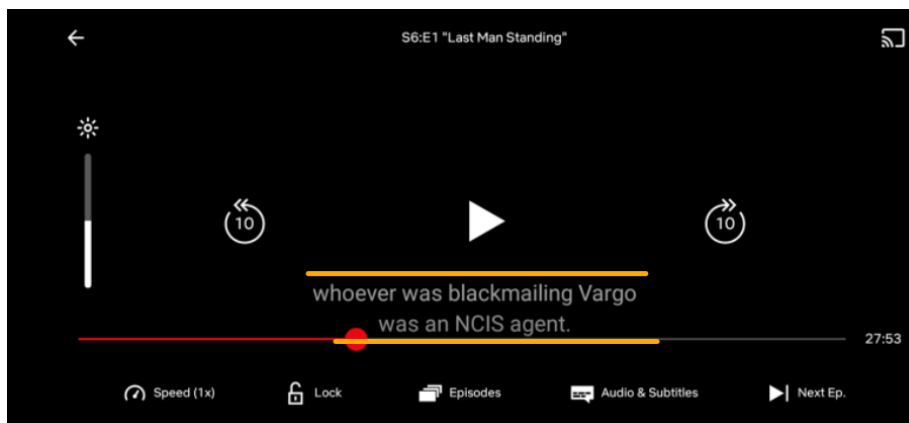


Figure 14 Accessibility in Netflix

As we can see, Netflix prioritize accessibility through closed caption as well as audio descriptions for their content for users who have visual and hearing impairments.

3.1.8 User Control:

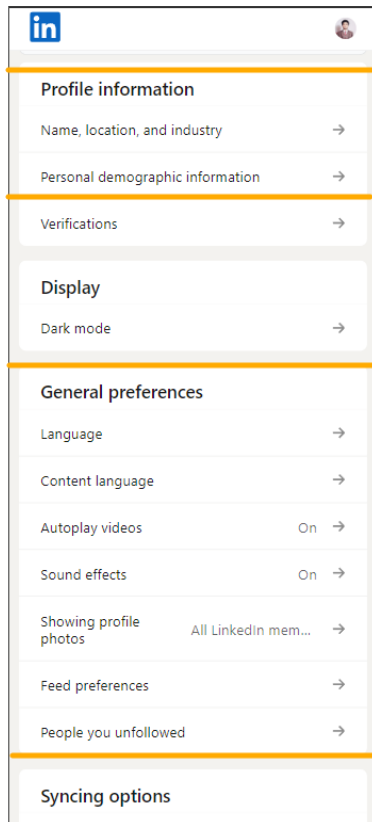
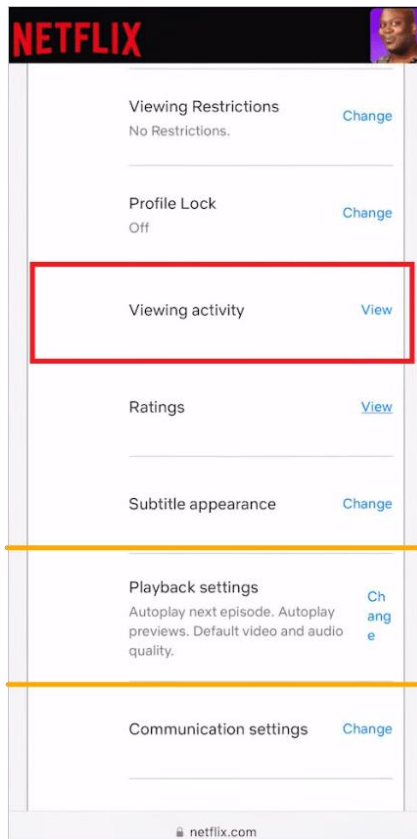


Figure 15 User Control in LinkedIn

As we can see, LinkedIn gives user to control profile settings, the feeds that the users prefer, they can see the people unfollowed which gives users control over their choices.



*Figure 16*User Control IN Netflix

As we can see, in Netflix too the user can have control over their viewing experience through adjusting playback settings, viewing activity and can create personalized profiles.

3.2 Nielsen's 10 Usability Heuristics:

3.2.1 Visibility of System Status:

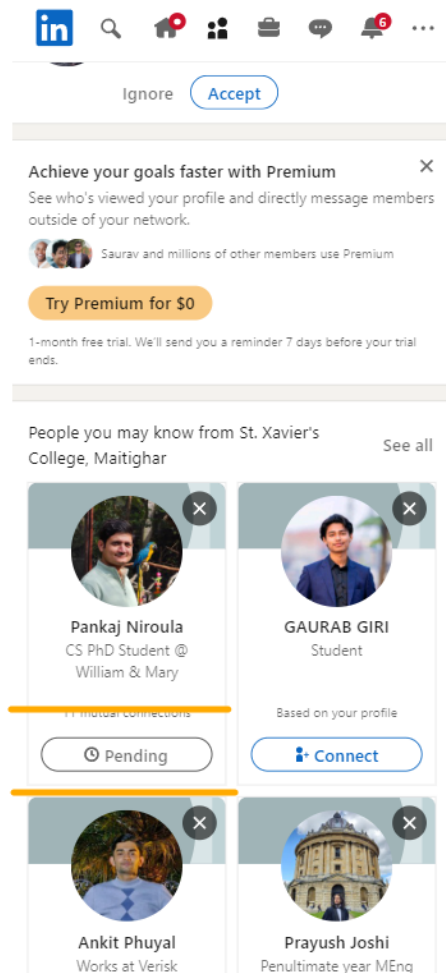


Figure 17 Visibility of System Status in LinkedIn

As we can see, in LinkedIn it provides a status indicator, pending until the request is declined or accepted.

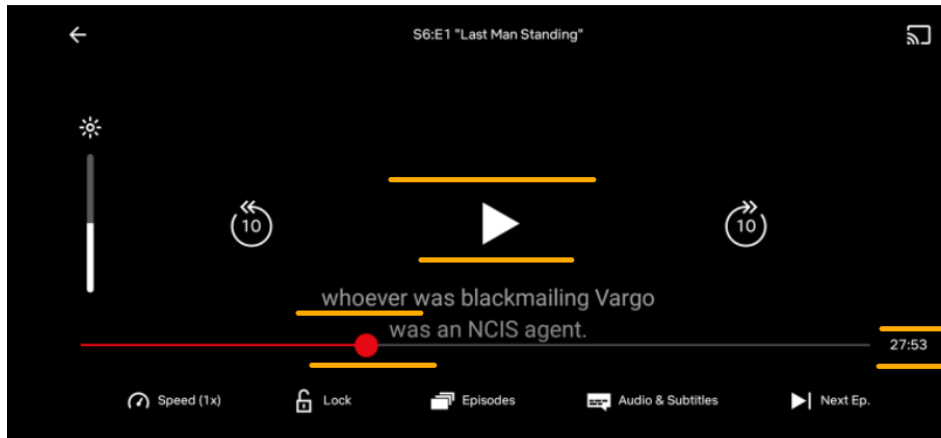


Figure 18 Visibility of System Status in Netflix

As we can see, in Netflix play, pause and the current playback time is provided.

3.2.2 Match between System and the Real World:

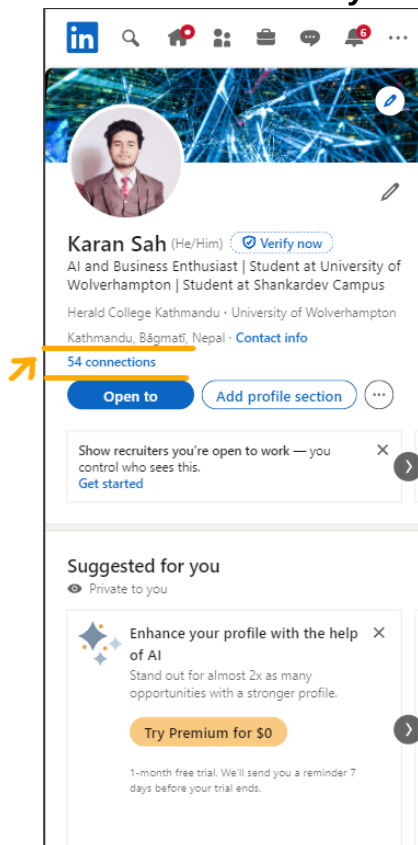


Figure 19 Match between System and the Real World in LinkedIn

As we can see, in LinkedIn connections for contacts, Inbox for messages are the user's real world understandings of the same thus making using LinkedIn easier.

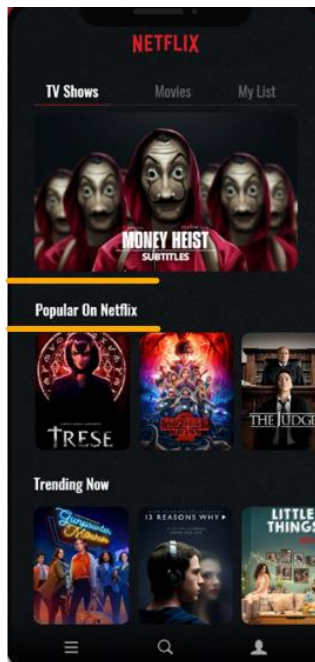


Figure 20 Match between System and the Real World in Netflix

As we can see Popular on Netflix and Genres mimic the real world experiences.

3.2.3 User Control and Freedom:

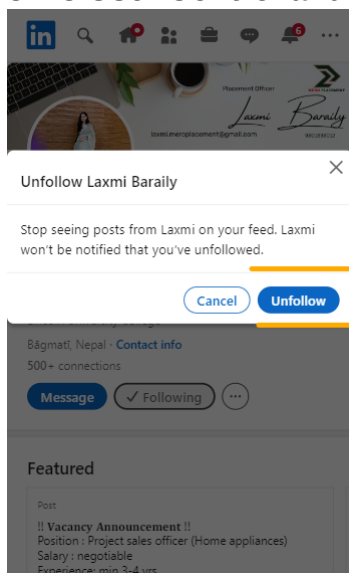


Figure 21 User Control and Freedom in LinkedIn

As we can see, in LinkedIn user can easily follow and unfollow any person giving user the control.

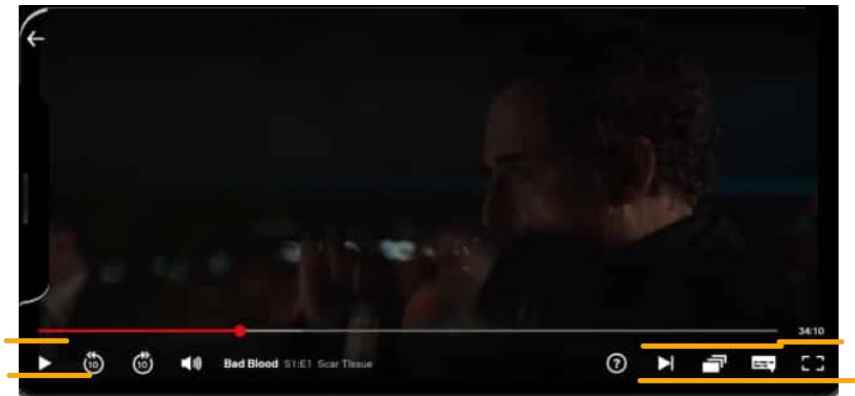


Figure 22 User Control and Freedom in Netflix

Netflix user can play, pause, press next, exit full screen mode giving them controls.

3.2.4 Consistency and Standards



Figure 23 Consistency and Standards in LinkedIn

Navigation menu are all place at the bottom for easy access and have consistency.

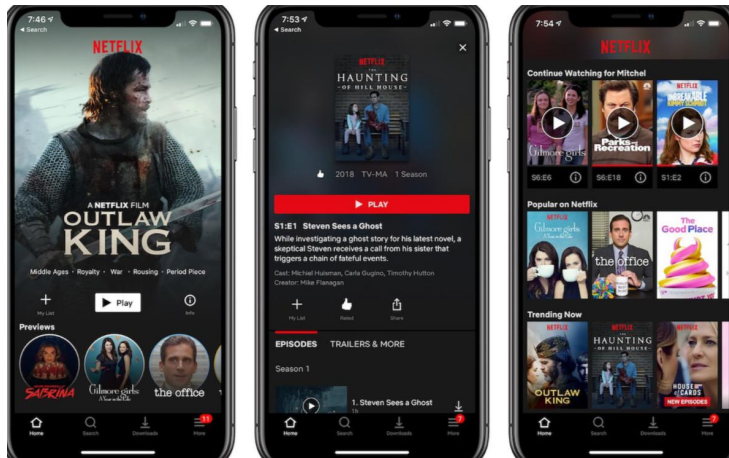


Figure 24 Consistency and Standards in Netflix

The layout and design of Netflix are consistent.

3.2.5 Error Prevention

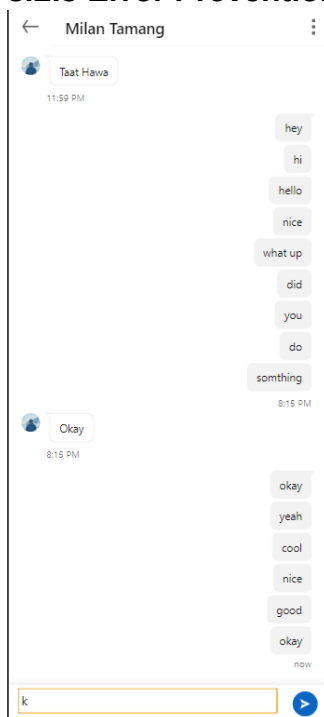


Figure 25 Error Prevention in LinkedIn

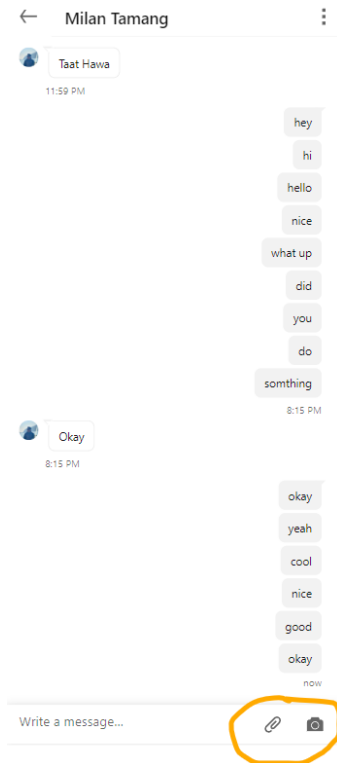


Figure 26 Error Prevention in LinkedIn 2

LinkedIn automatically removes the send button so that user don't send empty texts.



Figure 27Error Prevention in Netflix

Netflix uses clear icons and button for user's guide to prevent accidental touches.

3.2.6 Recognition rather than recall

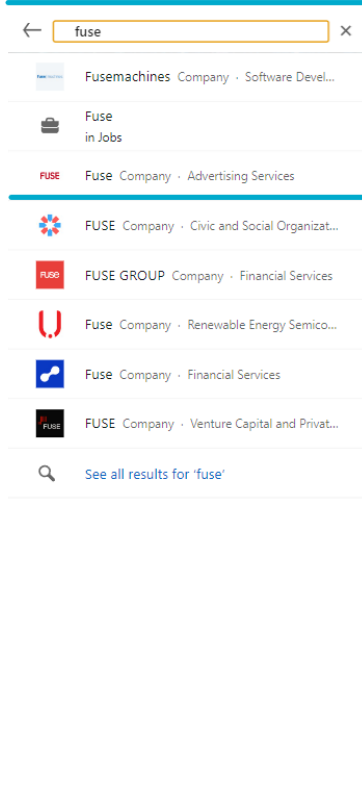


Figure 28 Recognition rather than recall in LinkedIn

LinkedIn provide suggestions to autocomplete during connections search which reduce the need for users to remember.



Figure 29 Recognition rather than recall in Netflix

Netflix remembers the past watched history, thus making it easier for user to continue.

3.2.7 Flexibility and efficiency of use

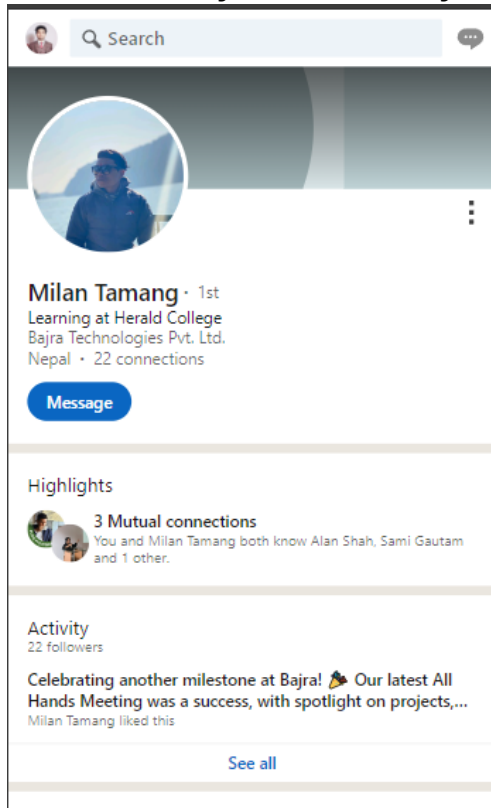


Figure 30 Flexibility and efficiency of use in LinkedIn

Without losing context, users can swiftly view the profile of their connections.

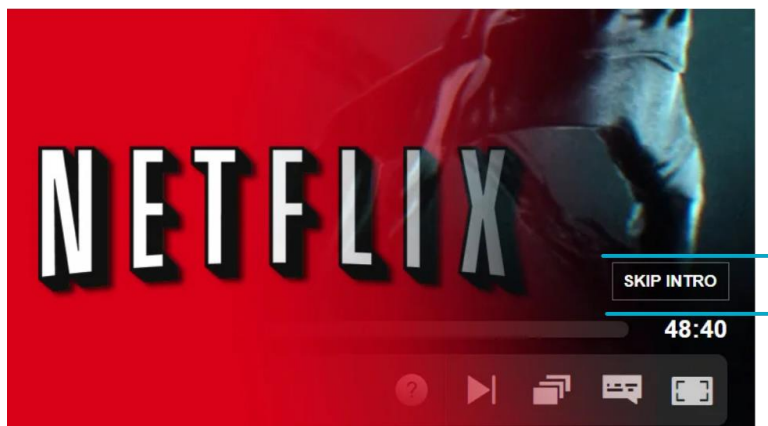


Figure 31 Flexibility and efficiency of use in Netflix

Netflix offers skip intro button to skip the opening credits for efficient viewing experience.

3.2.8 Aesthetic and Minimalist Design

As seen above both LinkedIn and Netflix have very simple and minimal design avoiding clutters thus making the app efficient to use.

3.2.9 Help Users Recognize, Diagnose and Recover from Errors

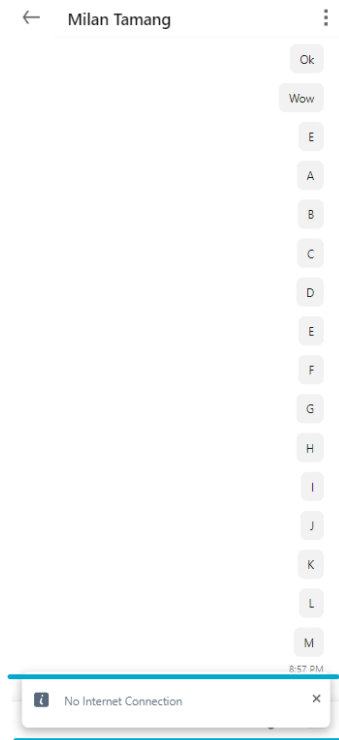


Figure 32 Help Users Recognize, Diagnose and Recover from Errors in LinkedIn

As the action to deliver message fails, no internet connection error shows informing them of the error in LinkedIn.

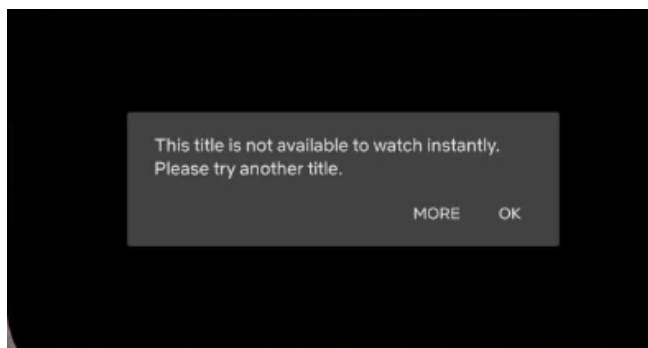


Figure 33 Help Users Recognize, Diagnose and Recover from Errors in Netflix

If there's an issue with playback please try again message is shown in Netflix suggesting users of the possible steps.

3.2.10 Help and Documentation

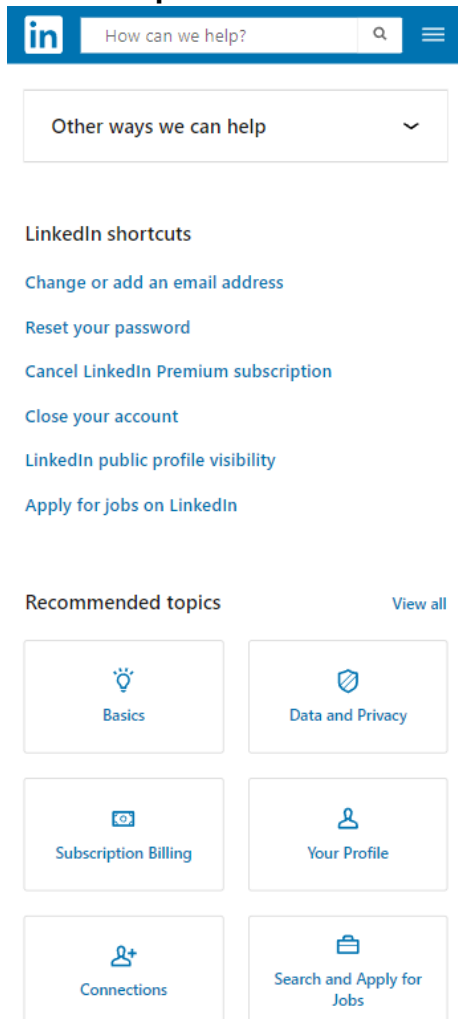


Figure 34 Help and Documentation in LinkedIn

There's help section in LinkedIn for the problems encountered.

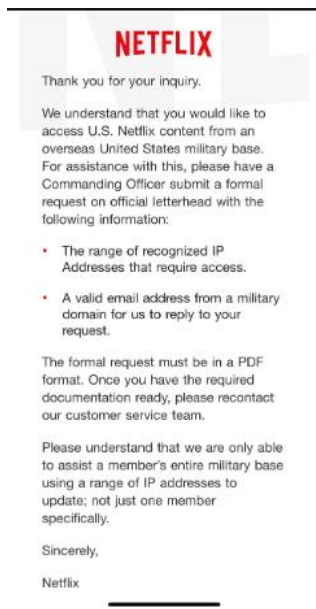


Figure 35Help and Documentation in Netflix

There are help section in Netflix which are great for inquiries too.

4. Conclusion

Based on the analysis of LinkedIn and Netflix against Shneiderman's "8 Golden Rules of Interface Design" and Nielsen's 10 Usability Heuristics, it is evident that both applications emphasize user experience by adhering to fundamental usability and interactive design principles. The key aspects highlighted are consistency, informative feedback, efficiency and shortcuts, and error handling.

The analysis demonstrates that these principles are highly applicable when designing an application for booking shows, as they help maintain consistency, provide feedback, handle errors, and empower users. These rules guide the design process towards a user-friendly and intuitive booking experience that meets or exceeds user expectations.

When designing my own interface, the points that should most influence the design are consistency in design language and interaction patterns, providing informative feedback throughout the booking process, implementing error prevention and clear error handling mechanisms, and empowering users with control over their actions. Incorporating these principles will create a user-friendly and efficient booking experience aligned with user expectations and best practices.

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Appendix

